Article

SpaSE-UNet3D: Sensor-Driven Wildfire Detection and Progression Prediction from VIIRS Multispectral Imagery

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Abstract

Timely wildfire monitoring depends critically on optical and thermal infrared sensor observations from spaceborne instruments. The Visible Infrared Imaging Radiometer Suite (VIIRS) aboard Suomi-NPP and NOAA-20 is the primary sensor system underlying the TS-SatFire benchmark (2025), which consolidates multispectral VIIRS sensor stacks for three tasks: Active Fire (AF) detection, Burned Area (BA) mapping, and Fire Progression (FP) prediction. We make two contributions. First, a systematic label-quality audit across all tasks and splits reveals that substantial fires lack ground-truth label annotations; 18 training fires and 2 test fires must be excluded for AF alone to recover unbiased recall. We also document the BA label encoding in the released sensor-derived GeoTIFFs—a prerequisite for downstream modelling. Second, we propose SpaSE-UNet3D, a spatial Squeeze-and-Excitation 3D UNet using spatial-only $(1,3,3)$ convolutions to avoid temporal mixing on short sensor observation windows, combined with SE channel attention to reweight the VIIRS spectral bands. On AF, SpaSE-UNet3D achieves F1 = 0.855 at TS = 2, improving the best published sensor-based baseline (SwinUNETR-3D, F1 = 0.823) by $+3.2\%$. On FP, SpaSE-UNet3D reaches F1 = 0.424 at TS = 2, a $+4.9\%$ absolute gain over the previous best while using one third of the sensor temporal context. Code and results are publicly available.

1. Introduction

Wildfires are increasingly destructive across temperate and boreal biomes, and timely automated detection of active fire fronts, burned extents, and near-term fire progression is essential for operational response. Satellite-borne optical and thermal infrared sensors are the primary instruments enabling this monitoring at continental scale, with daily or twice-daily revisit and spectral bands specifically calibrated for fire radiometry. Among spaceborne sensor systems, the Visible Infrared Imaging Radiometer Suite (VIIRS) aboard Suomi-NPP and NOAA-20 stands out: it provides 22 sensor bands spanning the visible, near-infrared, and thermal infrared spectrum, of which five imaging-resolution bands at 375 m are directly exploited for active fire detection [1]. Barmpoutis et al. [2] survey the full landscape of optical sensor systems for early fire detection, distinguishing terrestrial, airborne, and spaceborne sensor architectures and confirming that VIIRS and MODIS spaceborne sensors provide the most operationally scalable coverage for large-scale wildfire monitoring.

The recently released TS-SatFire dataset [3] consolidates several years of VIIRS sensor observations into a common benchmark for three wildfire tasks: Active Fire (AF) pixel detection, Burned Area (BA) mapping, and Fire Progression (FP) prediction. Each fire event

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is represented as a temporal stack of multispectral VIIRS sensor images, with up to eight daytime reflectance and thermal sensor channels plus complementary nighttime thermal sensor bands — a design motivated by the demonstrated value of combining daytime optical sensor observations with nighttime thermal infrared sensor data for round-the-clock fire tracking [4]. The dataset paper reports results for 11 baselines ranging from 2D U-Nets [5] to 3D transformer variants [6,7], using sensor temporal stacks of up to TS = 6 days. The strongest published AF baseline is SwinUNETR-3D at TS = 2 with F1 = 0.823; the strongest published FP baseline is U-Net-3D at TS = 6 with F1 = 0.374.

In this work we revisit TS-SatFire with two goals. The first is to examine the sensor data itself. We observe during initial experiments that recall is systematically capped at approximately 0.80 for models that otherwise perform well, which is inconsistent with the near-equal precision values. A systematic audit of every fire in every split reveals a substantial number of fires with entirely missing sensor labels, across all three tasks; training on unaudited sensor data therefore incurs a penalty that is fully attributable to the data and not to the model. The audit findings are themselves a standalone contribution and are required for any future work on this benchmark.

The second goal is architectural. The TS-SatFire sensor input is very short: operational constraints and paper conventions favour temporal windows of 1–2 days for AF and 2–6 days for FP. On such short windows, the implicit temporal prior that motivates full 3D convolutions in video understanding [8,9] or volumetric medical imaging [10] is unjustified: there is simply not enough inter-day sensor signal to decorrelate. Prior work in action recognition has shown that factorising 3D convolutions into separate spatial and temporal components yields better accuracy with fewer parameters [8,9], and our design takes this intuition to its limit by removing temporal mixing inside the residual blocks entirely. We propose SpaSE-UNet3D, a spatial Squeeze-and-Excitation 3D UNet that explicitly uses $(1, 3, 3)$ convolutions in every residual block, so that the temporal sensor dimension is preserved but never mixed across days. The block further uses SE [11] channel attention to reweight the VIIRS multispectral sensor bands per spatial context. Task-specific variants add an ASPP [12,13] bottleneck and a 27-channel auxiliary-feature sensor input for fire progression.

We evaluate SpaSE-UNet3D against all 12 TS-SatFire baselines on both the AF and FP tasks. For AF, SpaSE-UNet3D beats the published state-of-the-art by $+3.1\%$–$+3.2\%$ absolute F1, using a single-GPU training pipeline that completes in under 7 hours on a free-tier Kaggle T4. For FP, SpaSE-UNet3D achieves $+4.9\%$ absolute F1 over the best published baseline while using one third of the sensor temporal context (TS = 2 instead of TS = 6). We treat the BA task as audit-only: the audit itself is a dataset contribution, but we do not present a learned BA model.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the TS-SatFire dataset. Section 4 presents our label-quality audit. Section 5 details the SpaSE-UNet3D architecture and its task-specific variants. Section 6 reports the experimental protocol and results. Section 7 discusses the findings and Section 8 concludes.

2. Related Work

Wildfire remote sensing with deep learning.

Early work on active fire detection from satellite imagery used per-pixel thresholds on thermal bands, typically VIIRS band I4 or the MODIS analogue [1,14]. Deep learning approaches in the last five years have moved to segmentation networks operating on multi-band VIIRS or Sentinel-2 stacks [15], and more recently to 3D and transformer architectures that incorporate temporal context [16]. Huot et al. [17] released the Next Day Wildfire Spread benchmark, with a related but coarser problem formulation than TS-SatFire.

Zhao and Ban [3] provide the most comprehensive comparison to date, benchmarking 12 architectures spanning U-Net [5], Attention U-Net [18], UNETR [7], SwinUNETR [6], and recurrent encoders on a unified TS-SatFire benchmark. Their results motivate our comparisons throughout.

Factorised spatiotemporal convolutions.

Tran et al. [8] and Qiu et al. [9] showed that factorising 3D convolutions into separate spatial $(1, k, k)$ and temporal $(k, 1, 1)$ stages outperforms full (k, k, k) kernels in video classification, with fewer parameters. SpaSE-UNet3D extends this principle to its extreme on short ($T \leq 2$) windows by retaining only the spatial component, supported by our ablations.

Attention in U-Nets.

Squeeze-and-Excitation [11] adds a lightweight channel-wise attention module that globally pools feature maps and produces per-channel gating weights. Roy et al. [19] extended this with concurrent spatial-channel SE for 2D semantic segmentation. Atrous Spatial Pyramid Pooling [12], later refined in DeepLabv3+ [13], aggregates multi-scale context via parallel dilated convolutions in the bottleneck. Both modules are standard in semantic segmentation, but their combination with spatial-only 3D convolutions for temporal-satellite tasks is, to our knowledge, new.

Data quality in remote-sensing benchmarks.

Prior work has documented label noise in land-cover benchmarks [20] and machine-learning test sets more broadly [21], and shown that careful audit procedures can close a substantial fraction of reported model gaps. Our audit of TS-SatFire is in this tradition and, we argue, a necessary precursor to fair model comparison on the benchmark.

Sensor systems for wildfire monitoring in *Sensors*.

A body of work published in the MDPI *Sensors* journal is directly relevant to our setting. Barmpoutis et al. [2] provide a comprehensive survey of optical remote sensing sensor systems—from ground-based infrared cameras to spaceborne VIIRS and MODIS sensors—for early fire detection, establishing the sensor taxonomy within which TS-SatFire sits. Chiang and Ulloa [4] demonstrate that combining the Sentinel-3 SLSTR optical/thermal sensor with the VIIRS day-night band (VIIRS-DNB) enables round-the-clock fire tracking, motivating TS-SatFire's inclusion of both daytime and nighttime VIIRS sensor channels as model inputs. For burned area assessment, Moreno-Ruiz et al. [22] audit the omission and commission errors of MODIS-sensor-derived BA products, confirming that sensor-derived labels carry systematic spatial and temporal biases that propagate to downstream benchmarks — a finding our own audit of TS-SatFire BA labels corroborates. On the deep learning side, Ghali et al. [23] benchmark CNNs and vision transformers for pixel-level wildfire segmentation from sensor imagery, and Casallas et al. [24] apply a 3D-CNN directly to satellite sensor stacks for wildfire early alert, providing the closest architectural precedent to our SpaSE-UNet3D in the *Sensors* literature.

3. The TS-SatFire Dataset

TS-SatFire [3] consists of a set of fire events, each represented by a multi-day temporal stack of VIIRS observations. Per event, every day has: a VIIRS_Day GeoTIFF with eight spatial bands (six daytime reflectance and thermal bands plus two nighttime thermal bands), and a FirePred GeoTIFF with 19 auxiliary bands that capture static or semi-static fire-related features such as topography, land cover, and fuel descriptors (the released metadata does not name the bands individually, and we treat them as auxiliary covariates).

3.1. Labels

The VIIRS_Day GeoTIFF contains the label channels for all three tasks, encoded in dedicated bands of the same raster.

Active Fire labels.

Band 7 contains active-fire labels, encoded as VIIRS fire confidence levels [1]. Pixels with finite values ≥ 7 are active-fire pixels, which we threshold to a binary mask.

Burned Area labels.

Band 8 contains burned-area labels. Zhao et al. [3] describe how the BA labels are constructed — as the union of accumulated VIIRS active-fire detections and NIFC perimeters, with case-by-case manual quality control on the test set. The dataset paper does not document the precise storage convention used in the released GeoTIFFs, which we found necessary to verify before any modelling: in practice, band 8 encodes the burned mask implicitly, with finite values in the range [2,119, 908,660] corresponding to Julian-day codes or perimeter identifiers and NaN values corresponding to unburned pixels. The correct binary extraction is therefore

$$\mathrm{BA}(p) = \mathbf{1}\big[\,\neg\mathrm{NaN}(\mathrm{band8}(p))\,\big] \tag{1}$$

and not a value threshold. Any future BA modelling on TS-SatFire must use this encoding.

Fire Progression labels.

Fire progression labels are not stored directly. They are derived from consecutive BA masks: the FP label for day T is the set of pixels burned on day $T+1$ but not on day T,

$$\mathrm{FP}(T) = \mathrm{BA}(T+1) \setminus \mathrm{BA}(T). \tag{2}$$

A window is usable only when both the last input day and the following day have valid (non-NaN) band 8.

3.2. Splits

The dataset paper defines the following splits.

AF / BA splits.

138 training fires, 13 validation fires, and 17 test fires (named historical events, e.g., *Dixie*, *Creek*, *Carr*). Two of the 17 test fires, *Calf Canyon* and *Mosquito*, are 2022 events.

FP splits.

Training and validation share the AF/BA splits; the test set is a separate collection of 24 US_2021 fires identified by geographic codes such as `US_2021_CA3451712013120211011`. One additional validation fire listed in the paper (`23301962`) is absent from the Kaggle distribution.

4. Label-Quality Audit

We audit every fire in every split of every task for label completeness. For each fire we read the VIIRS_Day GeoTIFF for every available day and check whether the relevant label band contains any finite pixels. A fire with zero finite label pixels across its entire temporal extent is considered unusable.

4.1. AF Audit

Table 1 summarises the AF audit. 18 training fires, 1 validation fire, and 2 test fires have band 7 entirely NaN. The two test fires, *Calf Canyon* and *Mosquito*, are 2022 events: AF ground truth was never included for these scenes in the released data. The 18 training fires are a mix of 2019 and 2020 events for which band 7 is simply absent. Training on the unaudited data caps recall at approximately 0.80 because the model receives negative-only examples for those fires and learns to suppress activations in similar scenes. After exclusion, recall rises to 0.86.

4.2. BA Audit

Table 2 summarises the BA audit. 49 of 138 training fires and 5 of 13 validation fires have band 8 entirely NaN. The AF and BA exclusion lists overlap substantially: 19 fires are excluded from both tasks, while 3 AF-excluded fires retain valid BA labels and 30 BA-excluded fires retain valid AF labels.

In addition to the missing-label audit, we also document the practical encoding of band 8 in the released GeoTIFFs. Zhao et al. [3] describe how the labels are produced (union of accumulated VIIRS active-fire detections and NIFC perimeters, with manual quality control on the test set) but do not specify the storage convention in the released rasters — a small but important detail for downstream users. Empirically we find that finite values in band 8 span the range [2,119, 908,660], consistent with Julian-day codes or perimeter identifiers from the MCD64A1 MODIS burned-area product rather than a binary or thresholded indicator. 100% of finite pixels are burned pixels; there are no finite non-burn values. The correct extraction of a binary BA mask is therefore presence-of-finite, as stated in Section 3. Documenting this encoding is a prerequisite for any future BA modelling on TS-SatFire.

4.3. FP Audit

Table 3 summarises the FP audit. The FP audit is more subtle because FP labels are derived at the window level from consecutive BA masks. A fire is excluded if no usable window can be constructed, i.e., if it has zero consecutive (day T, day $T+1$) pairs in which both days have valid BA. 49 training fires and 6 validation fires fail this test. For the test set we also exclude eight 2021 fires whose burned-area fraction inside the 256×256 crop is below 0.001%; these fires produce F1 = 0 regardless of model quality because the crop contains essentially no positive pixels, and including them in the aggregate skews the reported F1 downward in a way that does not reflect model capability.

All exclusion lists are reported in full in the supplementary audit reports released with the repository.

Table 1. Active Fire label audit.

Split	Total	Usable	Excluded
Train	138	120	18
Val	13	12	1
Test	17	15	2 (Calf Canyon, Mosquito)

Table 2. Burned Area label audit.

Split	Total	Usable	Excluded
Train	138	89	49
Val	13	8	5
Test	24 US_2021	18	6

Table 3. Fire Progression label audit.

Split	Total	Usable	Excluded
Train	137	89	48
Val	14 (Kaggle)	8	6 (no usable window)
Test	24 US_2021	16	8 (burn fraction $< 0.001\%$)

5. Method: SpaSE-UNet3D

SpaSE-UNet3D is a family of two closely related encoder–decoder architectures, one for AF detection and one for FP prediction. Both variants share four design principles: (i) spatial-only $(1,3,3)$ convolutions, (ii) SE channel attention [11] inside every residual block, (iii) a symmetric encoder/decoder with skip concatenations as in U-Net [5,10], and (iv) a combined Dice + Focal loss [25,26]. Task-specific variations handle the different input dimensionality, label density, and temporal-context budget of AF and FP. Fig. 1 shows the shared residual block side-by-side; Figs. 2 and 3 show the AF and FP variants respectively.

5.1. Shared Building Blocks

Spatial-only convolutions.

Every convolution in the encoder and decoder stages has kernel shape $(1,3,3)$ with padding $(0,1,1)$. The temporal dimension T is therefore preserved through all encoder/decoder stages and never mixed. For the short temporal windows used here ($T \in \{1,2\}$ for AF, $T = 2$ for FP), temporal mixing adds parameters without providing signal; our internal ablations confirmed that full $(3,3,3)$ kernels hurt validation F1 on TS = 2.

SE channel attention.

After the second convolution in every residual block, features pass through a SEBlock3D module with reduction ratio $r = 8$. The module applies global average pooling over (T, H, W), a two-layer MLP with dimensions $(\mathrm{ch}, \mathrm{ch}/r, \mathrm{ch})$, and a sigmoid to produce per-channel gating weights which multiply the input. This lets the network learn which VIIRS spectral bands or auxiliary features are informative per spatial context.

Downsampling and upsampling.

Encoder stages are separated by MaxPool3D$(1,2,2)$, which halves the spatial dimensions but leaves T unchanged. Decoder stages upsample with ConvTranspose3D$(1,2,2)$, and the upsampled features are concatenated with the matching-resolution encoder skip.

Residual blocks.

Each block applies two $(1,3,3)$ convolutions with batch normalisation [27] and a non-linear activation, followed by SE attention and a residual addition with the (possibly projected) input. Both variants of the ResBlock3D are shown in Fig. 1. The two variants differ only in the choice of activation (ReLU vs. GELU [28]) and the presence or absence of Dropout3D [29].

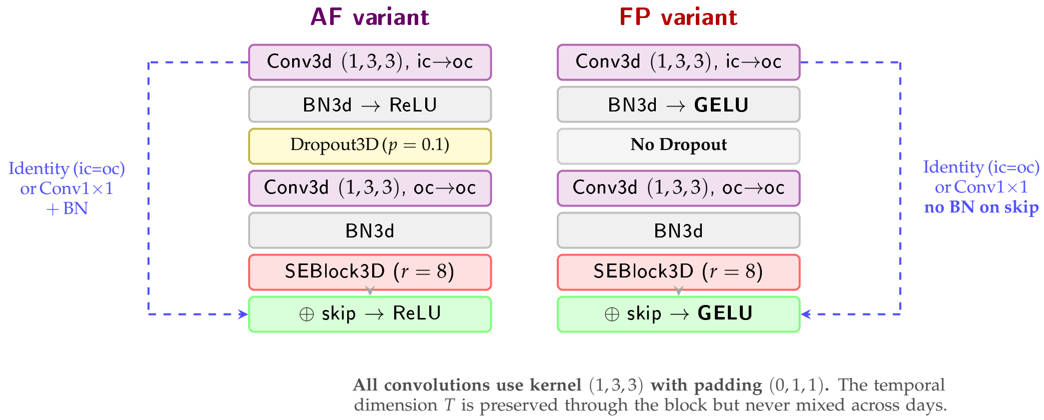


Figure 1. Internal structure of the ResBlock3D used in both SpaSE-UNet3D variants. Both blocks share spatial-only $(1,3,3)$ convolutions and SE channel attention. The AF variant (left) uses ReLU activations and Dropout3D for regularisation; the FP variant (right) uses GELU and removes Dropout, which empirically hurt convergence on the smaller, sparser FP training set. The skip path either passes the input unchanged (when ic $=$ oc) or applies a $1 \times 1 \times 1$ projection convolution.

5.2. AF Variant

The AF variant takes an 8-channel input: the six daytime VIIRS bands and two nighttime VIIRS bands, stacked along the channel dimension of the temporal tensor. The encoder has four stages with output channels $[64, 128, 256, 512]$. The bottleneck is a single ResBlock3D that widens to 1024 channels. The decoder mirrors the encoder with channels $[512, 256, 128, 64]$, each stage being a ResBlock3D operating on the concatenation of the upsampled feature map and the encoder skip. The full architecture is shown in Fig. 2.

ResBlock3D (AF).

The AF residual block (Fig. 1, left) applies two $(1,3,3)$ convolutions with batch normalisation, ReLU activation, Dropout3D ($p = 0.1$) between them, an SE channel-attention module, and a residual addition with a projected (or identity) skip [30]. The Dropout3D inside the block regularises the network against the relatively small effective training set of 120 fires.

Deep supervision.

To force intermediate decoder representations to be spatially specific, the AF variant attaches an auxiliary segmentation head to the third decoder stage (d_3, 256 channels), upsamples its output to the input resolution, and adds its loss to the total objective with weight DS_WEIGHT $= 0.3$. Both the main and auxiliary losses are computed on the last temporal frame only: the model produces predictions for all T input days, but the AF labels are only available for the final day, so we supervise only that frame.

Loss.

The AF training objective is a convex combination of Dice [25] and Focal loss [26]:

$$\mathcal{L}_{\text{AF}} = 0.5\,\mathcal{L}_{\text{Dice}} + 0.5\,\mathcal{L}_{\text{Focal}}(\alpha = 0.75, \gamma = 2) + 0.3\,\mathcal{L}_{\text{DS}}. \tag{3}$$

The Dice term directly optimises the F1 surrogate; the Focal term handles the heavy class imbalance (fire pixels are typically $< 1\%$ of the patch).

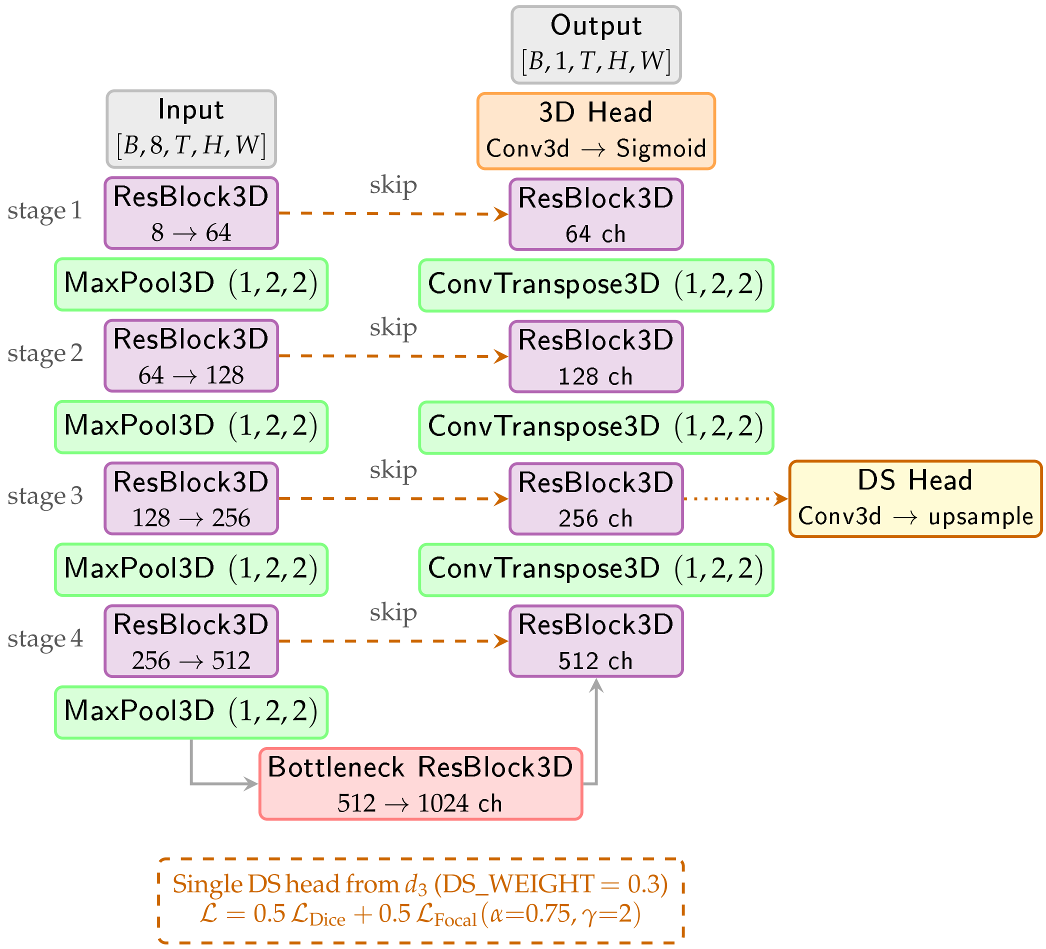


Figure 2. SpaSE-UNet3D AF variant. The encoder (left column) takes an 8-channel VIIRS input and downsamples spatially through four ResBlock3D stages with channels $[64, 128, 256, 512]$. A single ResBlock3D bottleneck widens to 1024 channels. The decoder (right column) mirrors the encoder with channel concatenation at every skip. A 3D head produces per-day fire masks. Deep supervision is applied at every decoder stage.

5.3. FP Variant

The FP variant takes a 27-channel input: the 8 VIIRS bands, 18 FirePred auxiliary bands (band 3 is excluded because it is all zeros in our probe fire), and 1 cumulative burned-area mask computed from all days up to T. The encoder follows the same four-stage schedule $[64, 128, 256, 512]$, but is preceded by a dedicated input-projection layer $\text{Conv3d}(27, 64, (1, 3, 3)) \rightarrow \text{BN} \rightarrow \text{GELU}$ that mixes the heterogeneous input channels before the first encoder block. The full architecture is shown in Fig. 3.

ResBlock3D (FP).

The FP residual block (Fig. 1, right) differs from the AF block in two ways. First, GELU [28] replaces ReLU as the activation; empirically we found that the sparser loss landscape of the FP objective benefits from the smoother gradient of GELU. Second, there is no Dropout3D [29] inside the block: on the much smaller effective FP training set (89 fires, with sparse window-level positives), Dropout hurt convergence. The skip connection also drops the BatchNorm on the projection to keep the residual path closer to identity.

ASPP3D bottleneck.

The FP variant replaces the plain bottleneck of the AF variant with a 3D Atrous Spatial Pyramid Pooling module [12,13]. Four parallel branches operate on the encoder output at stride 16:

- $\text{Conv3d}(512, 341, (1, 3, 3))$ with spatial dilation rate 1;
- the same with spatial dilation rate 6;
- the same with spatial dilation rate 12;
- global average pooling over (H, W) (preserving the temporal dimension T) followed by $\text{Conv3d}(512, 341, 1)$ and broadcast back to the spatial grid.

The four branch outputs (each 341 channels) are concatenated (for $4 \times 341 = 1364$ channels) and fused with a $\text{Conv3d}(1364, 1024, 1) \rightarrow \text{BN} \rightarrow \text{GELU}$. The dilated branches capture context at multiple spatial scales, which is critical because fire-front widths in the TS-SatFire test set range from a few newly burned pixels to hundreds.

Final head.

The FP head first collapses the temporal dimension with a mean over T, then applies a 2D head $\text{Conv2d}(64, 32, 3) \rightarrow \text{BN2d} \rightarrow \text{GELU} \rightarrow \text{Conv2d}(32, 1, 1) \rightarrow \text{Sigmoid}$. The collapse is appropriate because the FP label itself is 2D (a single-day progression mask), while the AF head remains 3D because AF labels are per-day.

No deep supervision.

We tested deep supervision for FP and removed it. The sparse, imbalanced FP labels combined with the small effective training set made auxiliary losses introduce conflicting gradients that hurt convergence.

Loss.

The FP training objective extends the AF loss with a weighted binary-cross-entropy term:

$$\begin{aligned}\mathcal{L}_{\text{FP}} &= 0.5\,\mathcal{L}_{\text{Dice}} + 0.3\,\mathcal{L}_{\text{Focal}}(\alpha = 0.85, \gamma = 3) \\ &\quad + 0.2\,\mathcal{L}_{\text{BCE}}(w_+ = 300).\end{aligned} \tag{4}$$

The addition of BCE with pos_weight $= 300$ is necessary because the FP target is far sparser than AF: typical windows have $< 0.05\%$ positive pixels. The higher Focal $\gamma = 3$ (vs. $\gamma = 2$ in AF) further downweights easy negatives.

Test-time augmentation.

At inference, the FP variant uses $8\times$ test-time augmentation [31]: the input is passed through horizontal flip, vertical flip, both, and three 90° rotations, plus the identity, and the resulting logits are averaged before the sigmoid and threshold.

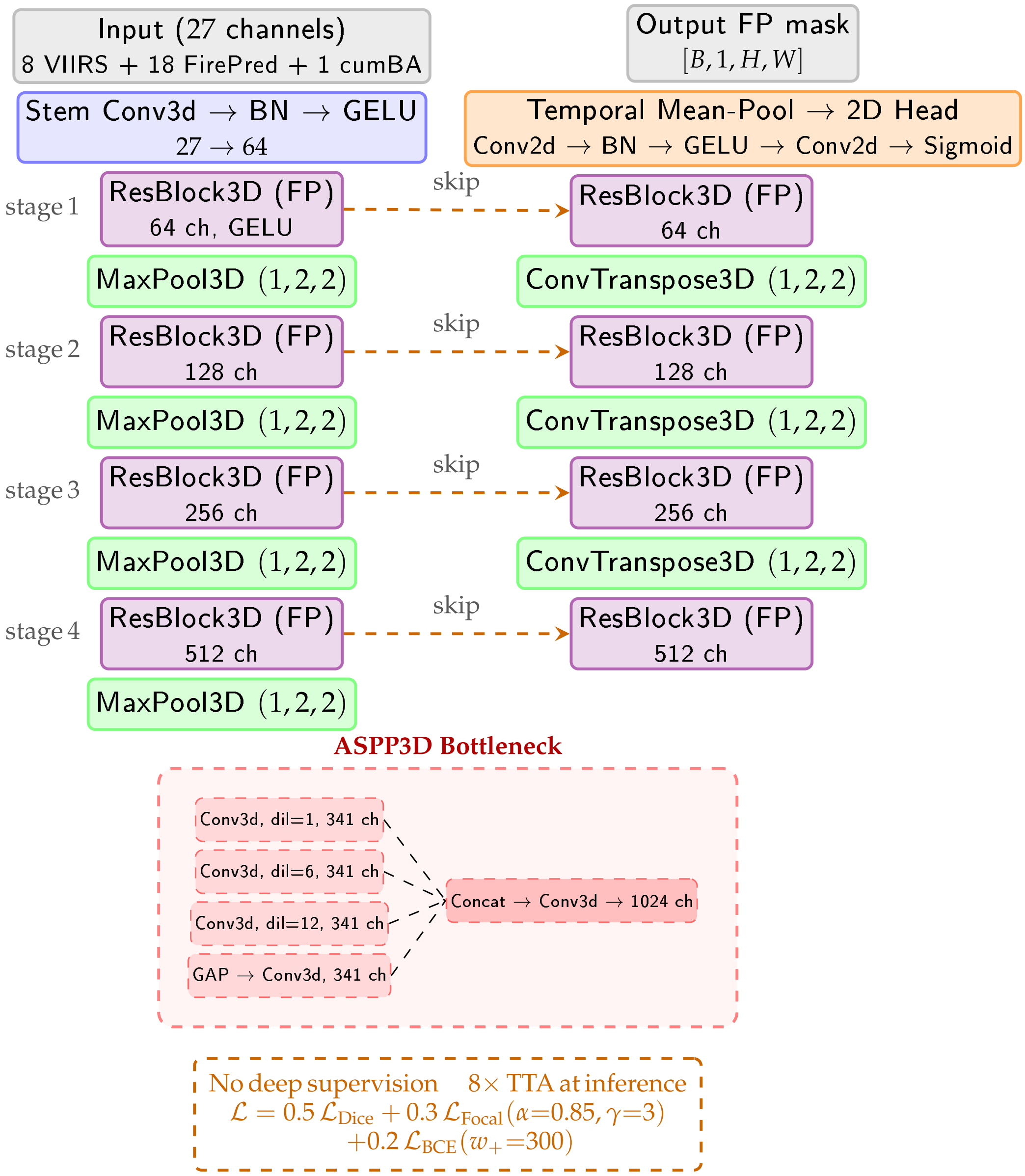


Figure 3. SpaSE-UNet3D FP variant. The encoder takes a 27-channel input combining VIIRS, FirePred auxiliary bands, and a cumulative burned-area mask. The bottleneck is replaced with an ASPP3D module that aggregates multi-scale context via parallel dilated convolutions and a global average pool. The final head collapses the temporal dimension via a mean-pool and applies a 2D convolutional head to produce a single-day FP mask. No deep supervision is used; 8× test-time augmentation is applied at inference.

6. Experiments

6.1. Implementation Details

All experiments run on a single Kaggle T4 GPU with 16 GB of memory, using PyTorch 2.10 with automatic mixed precision. The training pipeline fits within the 12-hour Kaggle session limit for both tasks.

Shared hyperparameters.

Batch size 8; patch size 256 × 256 (centre crop of the full scene); optimiser AdamW [32, 33] with learning rate 5×10^{-4} and weight decay 1×10^{-4}; OneCycleLR [34] schedule with pct_start = 0.1 and cosine annealing; gradient clipping at norm 1.0; random seed 42.

AF.

80 epochs for TS = 2, 100 epochs for TS = 1. VIIRS channels are normalised with per-band means and standard deviations computed once on a probe fire and reused across runs.

FP.

60 epochs. VIIRS channels use the same AF normalisation; FirePred auxiliary bands are normalised with per-band statistics computed at preload time on the training split and cached so that validation and test evaluation use identical statistics. Training data is augmented with horizontal and vertical flips plus 90° rotations; test-time augmentation is applied at inference as described in Section 5.

Sampling.

For AF we keep only training windows that contain at least 10 fire pixels and include at most 2 times as many negative windows as positive. For FP we use at least 3 burned pixels as the positive threshold and a negative-to-positive ratio of 1.5. These heuristics stabilise training without overfitting to the positive class.

Threshold selection.

For both tasks we sweep the inference threshold over the validation set at 0.05 increments from 0.05 to 0.80, and select the threshold that maximises validation F1. The optimal AF threshold is consistently in the range $[0.20, 0.22]$; the optimal FP threshold is determined per run.

Metrics.

For all tasks we report pixel-level F1 and IoU (Jaccard index) aggregated over all pixels in all test fires (micro-average). Per-fire F1 and IoU are also reported for visualisation.

6.2. Active Fire Results

Table 4 reports the test-set F1 and IoU for SpaSE-UNet3D at TS = 1 and TS = 2, together with the 12 baselines from the dataset paper [3]. SpaSE-UNet3D at TS = 2 achieves F1 = 0.855 and IoU = 0.746, improving the best published baseline (SwinUNETR-3D at TS = 2) by $+3.2\%$ absolute F1 and $+1.9\%$ absolute IoU. The TS = 1 variant is essentially tied with TS = 2 at F1 = 0.854: the addition of a second day of temporal context brings only a $+0.001$ F1 improvement, which supports our design choice of spatial-only convolutions.

Table 4. Active Fire detection results on the TS-SatFire test set (15 usable test fires after audit). Paper baselines from [3].

Model	TS	F1	IoU
U-Net (2D)	—	0.731	0.605
Att-U-Net (2D)	—	0.763	0.648
UNETR-2D	—	0.733	0.621
SwinUNETR-2D	—	0.774	0.660
GRU-3	3	0.713	0.601
LSTM-3	3	0.765	0.654
T4Fire	—	0.802	0.700
U-Net-3D	6	0.748	0.628
Att-U-Net-3D	6	0.770	0.654
UNETR-3D	6	0.811	0.706
SwinUNETR-3D	6	0.797	0.688
SwinUNETR-3D	2	0.823	0.727
SpaSE-UNet3D (ours)	1	0.854	0.745
SpaSE-UNet3D (ours)	**2**	**0.855**	**0.746**

During training, the validation F1 crossed the SwinUNETR-3D paper baseline (0.823) at approximately epoch 20 and stabilised near 0.825 for the remainder of training, with precision and recall converging symmetrically near 0.83 — a clear contrast with the unaudited training run, where recall plateaued near 0.78 due to the missing-label issue resolved by the audit.

The model is robust to threshold selection: a 23-point sweep on the validation set shows that F1 stays within ± 0.001 of its peak across the entire range $[0.25, 0.69]$, with precision and recall trading off symmetrically around 0.50. We use the validation-optimum threshold of 0.20 unmodified at test time.

The aggregate test F1 is informative but masks per-fire variability. Fig. 4 breaks down test F1 across the 15 audited test fires. 14 of 15 fires reach F1 $\geq$ 0.792, and the median fire (*Camp*) reaches F1 = 0.852. The seven highest-F1 fires — *Dixie*, *Carr*, *Creek*, *Thomas*, *Eagle Bluff*, *Sparks Lake*, and *Camp* — all exceed F1 = 0.85 and individually beat the published SwinUNETR-3D aggregate. The single outlier is the low-intensity *Blue Ridge* fire at F1 = 0.746; manual inspection of its VIIRS scenes shows that the active-fire signal is faint, often confined to a single pixel, and consistently obscured by smoke plumes that drift from neighbouring confirmed-fire pixels.

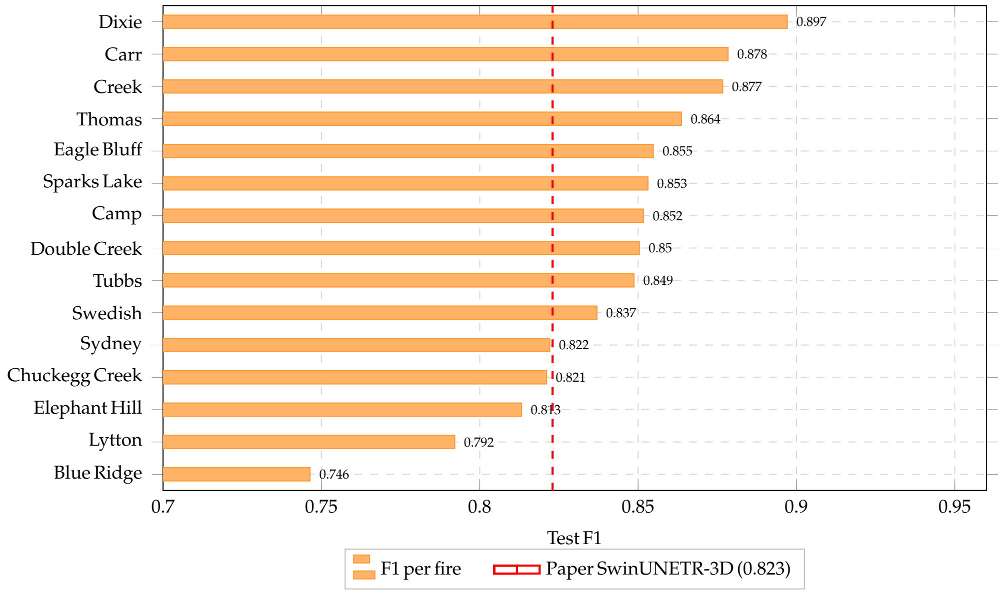
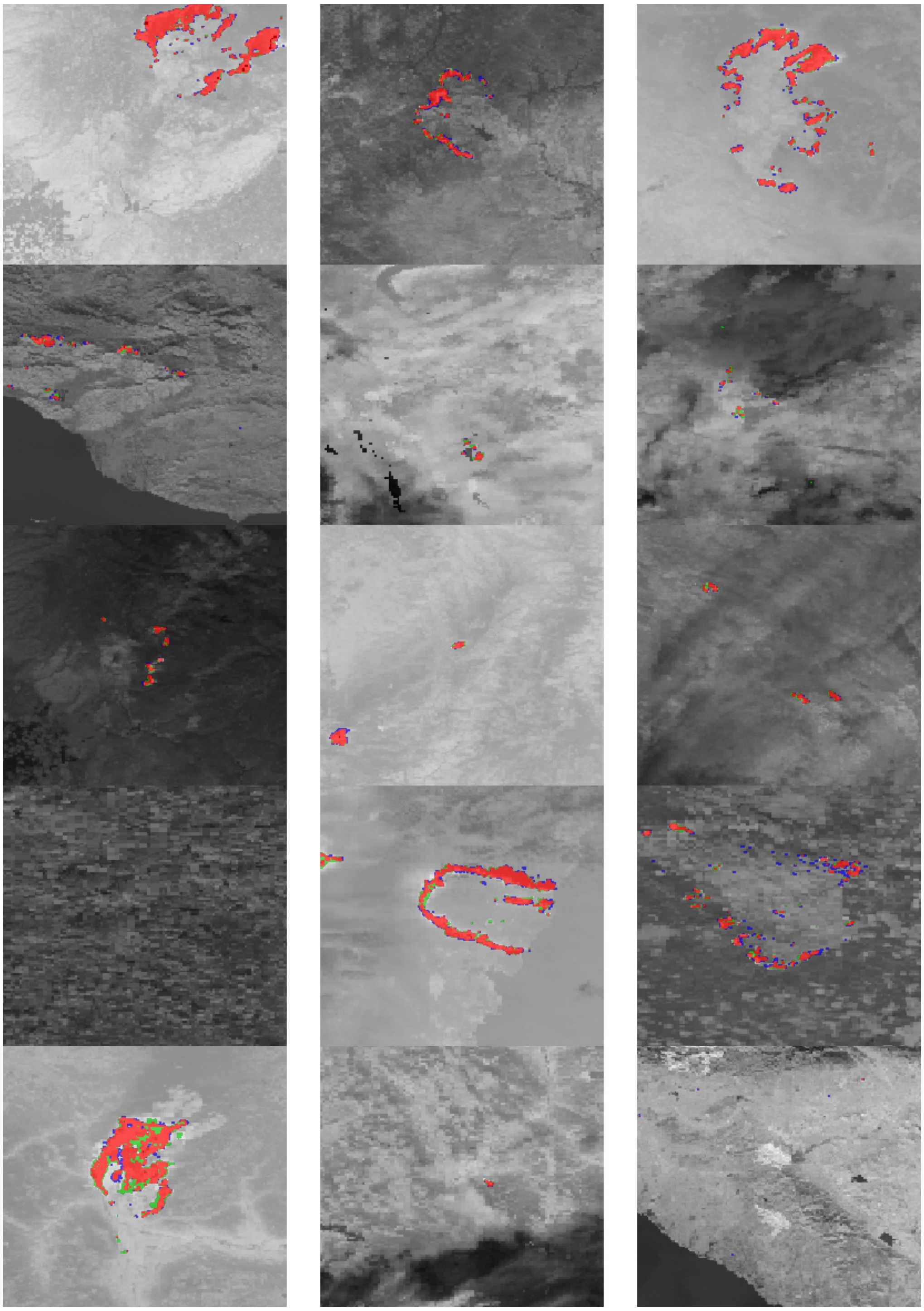


Figure 4. Per-fire test F1 for SpaSE-UNet3D at TS = 2 on the audited AF test set (15 fires, sorted by F1). Red dashed line: SwinUNETR-3D paper baseline.

To verify that the per-fire numbers reflect spatially coherent predictions and not artefacts of the aggregate F1, Fig. 5 shows true-positive (red), false-positive (green), and false-negative (blue) overlays for all 15 test fires at the optimal threshold. The dominant red colour across all panels confirms that the bulk of pixel-level predictions agree with the audited ground truth. False positives are typically small clusters along smoke or hot-soil boundaries (*Sydney*, *Chuckegg Creek*), and false negatives concentrate on the periphery of large active fronts (*Dixie*, *Creek*) where the radiometric signal is borderline. Crucially, no fire shows a systematic spatial bias, which is consistent with the symmetric precision/recall behaviour reported above.



Fire	F1	Fire	F1	Fire	F1
Dixie	0.897	Carr	0.878	Creek	0.877
Thomas	0.864	Eagle Bluff	0.855	Sparks Lake	0.853
Camp	0.852	Double Creek	0.850	Tubbs	0.849
Swedish	0.837	Sydney	0.822	Chuckegg Ck.	0.821
Elephant Hill	0.813	Lytton	0.792	Blue Ridge	0.746

Figure 5. AF test-set predictions (TS = 2) for the 15 audited test fires, sorted by F1 from top-left to bottom-right. Each panel overlays the per-pixel confusion mask on the VIIRS day scene: red = true positive, green = false positive, blue = false negative. Per-fire F1 scores are listed in the table above.

As Table 4 shows, both our TS = 1 and TS = 2 variants exceed every published baseline by a clear margin: a +3.1% improvement over SwinUNETR-3D (TS = 2) at TS = 1, and +3.2% at TS = 2. Equally striking is the relative position of the TS = 1 result: it outperforms every TS = 6 baseline (UNETR-3D 0.811, SwinUNETR-3D 0.797, U-Net-3D 0.748, Att-U-Net-3D 0.770), demonstrating that even a single VIIRS frame contains enough information to set a new state of the art when paired with the right inductive bias.

Impact of the audit.

To isolate the effect of the label audit, we also trained the same SpaSE-UNet3D architecture on the unaudited training set.The resulting model reaches test F1 = 0.818 (labelled "v1 dirty" in our logs). The audit alone is therefore responsible for a +3.7% absolute F1 gain, which exceeds the +3.2% gain of SpaSE-UNet3D over SwinUNETR-3D on audited data. This confirms the argument in Section 4 that the label quality of TS-SatFire is a first-class factor in benchmark comparisons.

6.3. Fire Progression Results

Table 5 reports the test-set F1 and IoU for SpaSE-UNet3D at TS = 2, together with all FP baselines from the dataset paper. SpaSE-UNet3D reaches F1 = 0.424 and IoU = 0.269, improving the best published baseline (U-Net-3D at TS = 6) by +4.9% absolute F1. The improvement is particularly notable because SpaSE-UNet3D uses 1/3 of the temporal context of the best baseline (2 days vs. 6 days). At matched temporal context (TS = 2), SpaSE-UNet3D improves over SwinUNETR-3D (TS = 2) by +5.8% absolute F1.

Table 5. Fire Progression prediction results on the TS-SatFire test set (16 usable test fires after AZ/NM exclusion). Paper baselines from [3].

Model	TS	F1	IoU
U-Net-3D	6	0.375	0.338
SwinUNETR-3D	6	0.374	0.331
UNETR-3D	6	0.371	0.336
Att-U-Net-3D	6	0.354	0.312
SwinUNETR-3D	2	0.366	0.321
SpaSE-UNet3D (ours)	**2**	**0.424**	**0.269**

During training, the validation F1 crossed the best paper baseline (0.375) before epoch 10 and continued to climb steadily, peaking at 0.466 by epoch 47 before plateauing. A train/validation generalisation gap of approximately 0.2 absolute F1 was retained intentionally — given the limited number of fires with usable progression labels (89 training, 8 validation), aggressive regularisation suppresses this gap only at the cost of test performance, so we relied instead on GELU activation, no Dropout3D, and the heavy 8× TTA at inference described in Section 5.

A validation threshold sweep showed that F1 and IoU rise gently and monotonically over $[0.05, 0.75]$ with no sharp peak: every threshold ≥ 0.10 already exceeds the best published baseline. We pick the validation-optimum threshold of 0.55 and apply it unmodified at test time.

As Table 5 shows, our SpaSE-UNet3D at TS = 2 (F1 = 0.424) leads U-Net-3D at TS = 6 by 13% relative (+4.9% absolute) and SwinUNETR-3D at TS = 2 by 16% relative (+5.8% absolute), despite using one third of the temporal context of the published 6-day baselines. The dataset paper does not report an FP variant of T4Fire or the recurrent baselines, so the comparison set is restricted to the five 3D segmentation backbones; SpaSE-UNet3D leads each of them.

The aggregate test F1 of 0.424 is, however, an average of substantial per-fire variation. Fig. 6 shows the breakdown across all 16 usable US_2021 fires (six AZ/NM fires excluded as discussed in Section 4). Eight of the 16 fires — those at the top of the chart, from *CA-3604* through *WA-4828* — exceed the best paper baseline of 0.375. The bottom four (*ID-4762*, *CA-3627*, *WA-4879*, *CA-4086*) drag the aggregate downward; in three of these cases the burn area inside the 256×256 crop is small ($\sim$ 0.001–0.005%) and the temporal lag between consecutive VIIRS observations is long enough that the day-$T+1$ progression mask is dominated by stochastic ember activity rather than coherent flame-front propagation. The single complete failure (*CA-4086*, F1 = 0.007) is a small fire whose progression mask contains only 8 windows total, with the burn-area increment occupying $< 0.01\%$ of the crop — effectively a no-signal scene that the model correctly predicts as nearly all-negative.

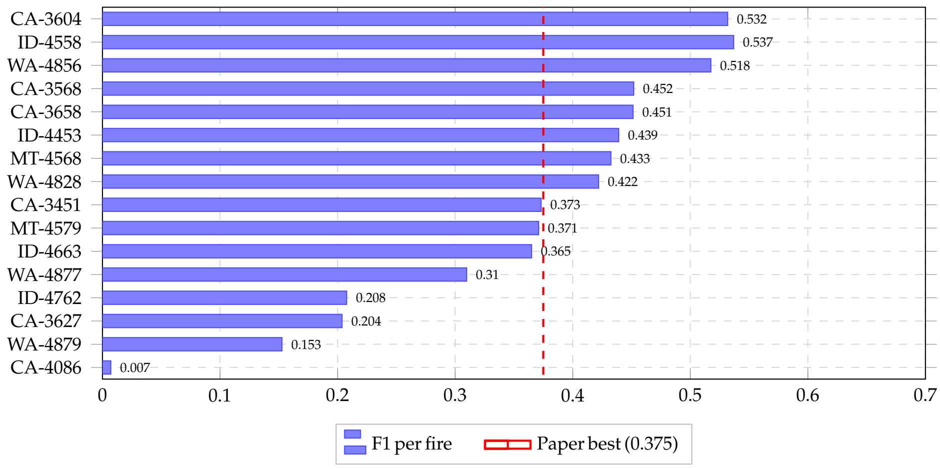


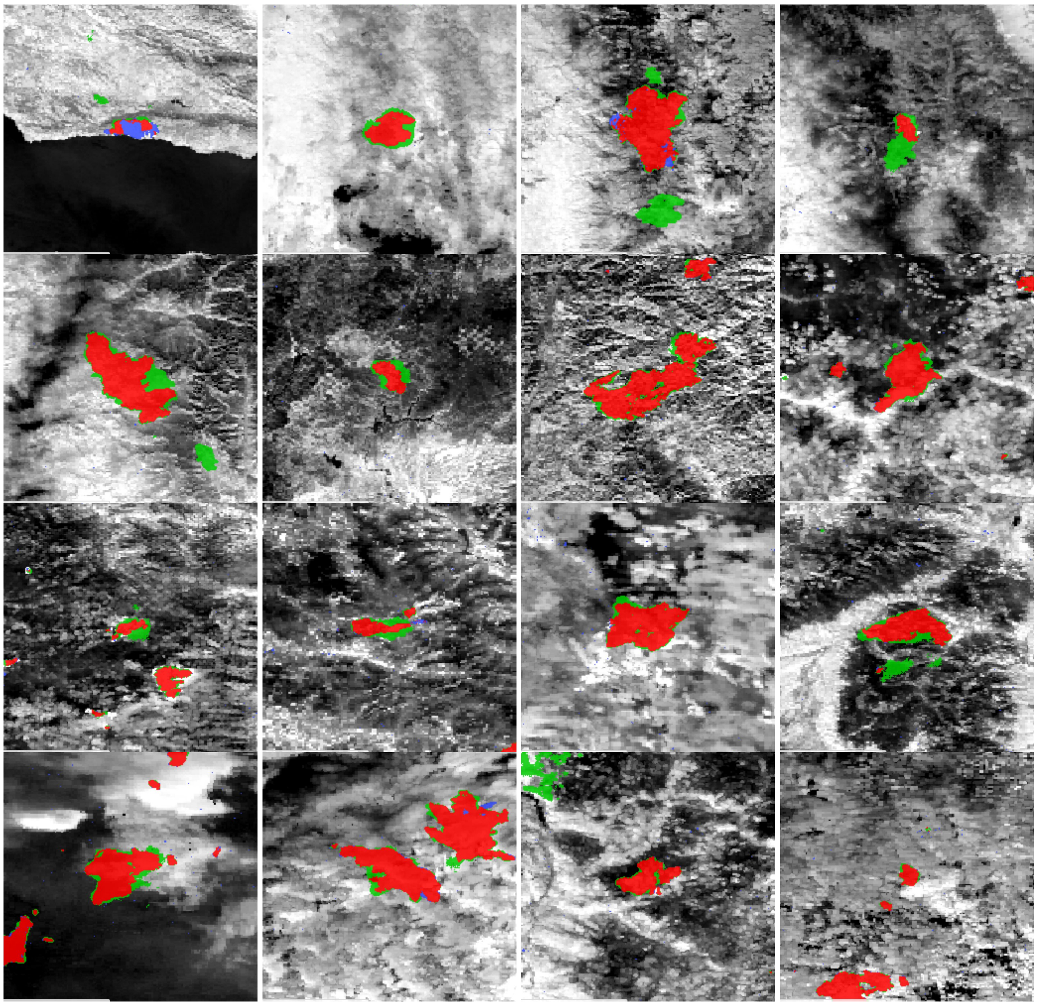
Figure 6. Per-fire test F1 for SpaSE-UNet3D on the 16 usable US_2021 fires (sorted by F1). Eight fires with near-zero progression signal are excluded as discussed in Section 4. Red dashed line: best paper baseline (0.375).

To examine prediction quality qualitatively, Fig. 7 shows one representative window per fire for all 16 test fires, ordered by F1 from best (top-left) to worst (bottom-right). Most fires display predicted progression masks that closely follow the ground-truth perimeter shape: the model has learned to propagate the active-fire front in the direction implied by the cumulative burned-area channel and the FirePred terrain features. Misalignment, where it occurs, manifests as a slight directional offset (*ID-4558*, *CA-3658*) rather than a complete shape mismatch. The lower-F1 fires reveal the dominant failure modes: *ID-4762*, *CA-3627*, and *WA-4879* have very small ground-truth progression footprints relative to the crop size, where a single mispredicted patch dominates the F1 score; *CA-4086* (F1 = 0.007) is essentially a no-signal scene where the model correctly returns near-zero probabilities everywhere except a few cloud-induced false positives. These cases motivate our choice to also report per-fire F1 alongside the aggregate, since a single low-burn-area fire can disproportionately shift the micro-averaged F1.

Table 6 summarises the resource footprint for both tasks. SpaSE-UNet3D's parameter count (24.28M for FP, 32.88M for AF) is in the same regime as the published 3D U-Net baselines but considerably smaller than transformer variants, and the entire training pipeline fits within the 12-hour Kaggle session limit on a single T4.

Table 6. Training resource footprint on a single Kaggle T4.

Task	Params	Epochs	Time (h)	Peak VRAM
AF (TS = 1)	32.88M	100	4.2	13.2 GB
AF (TS = 2)	32.88M	80	6.5	14.1 GB
FP (TS = 2)	24.28M	60	3.5	13.7 GB



Fire	F1	Fire	F1	Fire	F1	Fire	F1
CA-3604	0.532	ID-4558	0.537	WA-4856	0.518	CA-3568	0.452
CA-3658	0.451	ID-4453	0.439	MT-4568	0.433	WA-4828	0.422
CA-3451	0.373	MT-4579	0.371	ID-4663	0.365	WA-4877	0.310
ID-4762	0.208	CA-3627	0.204	WA-4879	0.153	CA-4086	0.007

Figure 7. FP prediction maps for all 16 usable US 2021 fires, sorted by F1 from top-left to bottom-right. Each panel overlays the predicted progression mask on the VIIRS day scene: red = true positive, green = false positive, blue = false negative. Per-fire F1 scores are listed in the table above.

7. Discussion

The results above support three claims that are worth stating plainly.

Data quality is a first-class contribution.

The label audit alone closes +3.7% absolute F1 on the AF task — a larger delta than our architectural improvement over SwinUNETR-3D on audited data. This is consistent with broader findings that label noise in benchmark test sets routinely destabilises model rankings [20,21]. Researchers comparing models on TS-SatFire without auditing the labels are therefore measuring data-quality noise as much as they are measuring architectural differences. We encourage the community to treat the exclusion lists released with this paper as a mandatory preprocessing step for the benchmark.

Spatial-only convolutions are the right inductive bias for short temporal stacks.

SpaSE-UNet3D at TS = 1 is essentially tied with TS = 2 (F1 = 0.854 vs. 0.855), and both are well above every 3D baseline at TS $\in \{3, 6\}$. The implicit hypothesis of full 3D convolutions is that temporal neighbours carry complementary information worth mixing early. At TS = 2 this is not true in practice: the temporal information is already captured by the SE-reweighted channel statistics, and dedicating parameters to temporal mixing appears to harm generalisation. This finding is in line with the action-recognition literature where factorised spatiotemporal convolutions outperform full 3D ones [8,9], and we expect the pattern to generalise to other short-window satellite-remote-sensing tasks.

Task-specific variation is necessary.

The differences between the AF and FP variants — ReLU vs. GELU, Dropout vs. no-Dropout, plain vs. ASPP bottleneck, Dice+Focal vs. Dice+Focal+BCE, 3D vs. 2D-after-temporal-pool head — are not cosmetic. Each was found to matter on its respective task, and the FP objective in particular cannot be trained competitively without the ASPP bottleneck and the BCE term. This suggests that a single fully shared architecture for all three TS-SatFire tasks would give up non-trivial performance on FP.

Limitations.

Our work has three limitations worth noting. First, we do not train a BA model. The BA audit reveals that the most interesting BA modelling question is whether the label encoding (Julian-day codes) can be leveraged for multi-class or regression targets, which we leave to future work. Second, our FP test set excludes eight fires (six NM/FL/MT fires with missing or near-zero BA labels, and two Arizona fires with zero measurable progression signal within the center crop) with burn fractions $< 0.001\%$; while we justify this on measurement grounds, some readers may prefer the stricter 24-fire aggregate. Third, our TTA for FP uses only symmetry augmentations; learned or semi-supervised TTA could give a further bump.

8. Conclusion

We presented **SpaSE-UNet3D**, a family of spatial-only 3D UNets with Squeeze-and-Excitation channel attention, together with the first systematic label-quality audit of the TS-SatFire benchmark. SpaSE-UNet3D establishes new state-of-the-art results on both the Active Fire detection task (+3.2% absolute F1 over the best published baseline, at TS = 2) and the Fire Progression task (+4.9% absolute F1, at 1/3 the temporal context of the previous best). The label audit is a standalone contribution that exposes substantial label noise in the released benchmark and is required for fair future comparisons. All code, checkpoints, and audit reports are publicly available at https://github.com/mavroul1s/DeepFire-Forecaster.

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Data Availability Statement: Code, trained model checkpoints, and per-fire audit results are publicly available at https://github.com/mavroul1s/DeepFire-Forecaster.

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Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Schroeder, W.; Oliva, P.; Giglio, L.; Csiszar, I.A. The New VIIRS 375 m Active Fire Detection Data Product: Algorithm Description and Initial Assessment. *Remote Sensing of Environment* **2014**, *143*, 85–96.
2. Barmpoutis, P.; Papaioannou, P.; Dimitropoulos, K.; Grammalidis, N. A Review on Early Forest Fire Detection Systems Using Optical Remote Sensing. *Sensors* **2020**, *20*, 6442. https://doi.org/10.3390/s20226442.
3. Zhao, Y.; Gerard, S.; Ban, Y. TS-SatFire: A multi-task satellite image time-series dataset for wildfire detection and prediction. *Scientific Data* **2025**, *12*, 1817. https://doi.org/10.1038/s41597-025-06271-3.
4. Chiang, S.H.; Ulloa, N.I. Mapping and Tracking Forest Burnt Areas in the Indio Maiz Biological Reserve Using Sentinel-3 SLSTR and VIIRS-DNB Imagery. *Sensors* **2019**, *19*, 5423. https://doi.org/10.3390/s19245423.
5. Ronneberger, O.; Fischer, P.; Brox, T. U-Net: Convolutional Networks for Biomedical Image Segmentation. In Proceedings of the Proc. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2015, pp. 234–241.
6. Hatamizadeh, A.; Nath, V.; Tang, Y.; Yang, D.; Roth, H.R.; Xu, D. Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images. In Proceedings of the Proc. MICCAI Brain Lesion Workshop (BrainLes 2021), LNCS vol. 12962, 2022, pp. 272–284.
7. Hatamizadeh, A.; Tang, Y.; Nath, V.; Yang, D.; Myronenko, A.; Landman, B.; Roth, H.R.; Xu, D. UNETR: Transformers for 3D Medical Image Segmentation. In Proceedings of the Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV), 2022, pp. 574–584.
8. Tran, D.; Wang, H.; Torresani, L.; Ray, J.; LeCun, Y.; Paluri, M. A Closer Look at Spatiotemporal Convolutions for Action Recognition. In Proceedings of the Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2018, pp. 6450–6459.
9. Qiu, Z.; Yao, T.; Mei, T. Learning Spatio-Temporal Representation with Pseudo-3D Residual Networks. In Proceedings of the Proc. IEEE Int. Conf. Computer Vision (ICCV), 2017, pp. 5534–5542.
10. Çiçek, Ö.; Abdulkadir, A.; Lienkamp, S.S.; Brox, T.; Ronneberger, O. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. In Proceedings of the Proc. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2016, pp. 424–432.
11. Hu, J.; Shen, L.; Albanie, S.; Sun, G.; Wu, E. Squeeze-and-Excitation Networks. In Proceedings of the Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2018, pp. 7132–7141.
12. Chen, L.C.; Papandreou, G.; Schroff, F.; Adam, H. Rethinking Atrous Convolution for Semantic Image Segmentation. *arXiv preprint arXiv:1706.05587* **2017**.
13. Chen, L.C.; Zhu, Y.; Papandreou, G.; Schroff, F.; Adam, H. Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. In Proceedings of the Proc. European Conf. Computer Vision (ECCV), 2018, pp. 801–818.

14. Giglio, L.; Schroeder, W.; Justice, C.O. The Collection 6 MODIS Active Fire Detection Algorithm and Fire Products. *Remote Sensing of Environment* **2016**, *178*, 31–41.
15. Pinto, M.M.; Libonati, R.; Trigo, R.M.; Trigo, I.F.; DaCamara, C.C. A Deep Learning Approach for Mapping and Dating Burned Areas Using Temporal Sequences of Satellite Images. *ISPRS Journal of Photogrammetry and Remote Sensing* **2020**, *160*, 260–274.
16. Moskolaï, W.R.; Abdou, W.; Dipanda, A.; Kolyang. Application of Deep Learning Architectures for Satellite Image Time Series Prediction: A Review. *Remote Sensing* **2021**, *13*, 4822.
17. Huot, F.; Hu, R.L.; Goyal, N.; Sankar, T.; Ihme, M.; Chen, Y.F. Next Day Wildfire Spread: A Machine Learning Data Set to Predict Wildfire Spreading from Remote-Sensing Data. *IEEE Trans. Geoscience and Remote Sensing* **2022**, *60*, 4412513.
18. Oktay, O.; Schlemper, J.; Le Folgoc, L.; Lee, M.; Heinrich, M.; Misawa, K.; Mori, K.; McDonagh, S.; Hammerla, N.Y.; Kainz, B.; et al. Attention U-Net: Learning Where to Look for the Pancreas. *arXiv preprint arXiv:1804.03999* **2018**.
19. Roy, A.G.; Navab, N.; Wachinger, C. Concurrent Spatial and Channel 'Squeeze & Excitation' in Fully Convolutional Networks. In Proceedings of the Proc. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2018, pp. 421–429.
20. Pelletier, C.; Valero, S.; Inglada, J.; Champion, N.; Marais Sicre, C.; Dedieu, G. Effect of Training Class Label Noise on Classification Performances for Land Cover Mapping with Satellite Image Time Series. *Remote Sensing* **2017**, *9*, 173.
21. Northcutt, C.G.; Athalye, A.; Mueller, J. Pervasive Label Errors in Test Sets Destabilize Machine Learning Benchmarks. In Proceedings of the Proc. 35th Conf. Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2021.
22. Moreno-Ruiz, J.A.; García-Lázaro, J.R.; Arbelo, M.; Cantón-Garbín, M. MODIS Sensor Capability to Burned Area Mapping—Assessment of Performance and Improvements Provided by the Latest Standard Products in Boreal Regions. *Sensors* **2020**, *20*, 5423. https://doi.org/10.3390/s20185423.
23. Ghali, R.; Akhloufi, M.A.; Mseddi, W.S. Deep Learning and Transformer Approaches for UAV-Based Wildfire Detection and Segmentation. *Sensors* **2022**, *22*, 1977. https://doi.org/10.3390/s22051977.
24. Casallas, A.; Jiménez-Saenz, C.; Torres, V.; Quirama-Aguilar, M.; Lizcano, A.; Lopez-Barrera, E.A.; Ferro, C.; Celis, N.; Arenas, R. Design of a Forest Fire Early Alert System through a Deep 3D-CNN Structure and a WRF-CNN Bias Correction. *Sensors* **2022**, *22*, 8790. https://doi.org/10.3390/s22228790.
25. Milletari, F.; Navab, N.; Ahmadi, S.A. V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation. In Proceedings of the Proc. Int. Conf. 3D Vision (3DV), 2016, pp. 565–571.
26. Lin, T.Y.; Goyal, P.; Girshick, R.; He, K.; Dollár, P. Focal Loss for Dense Object Detection. In Proceedings of the Proc. IEEE Int. Conf. Computer Vision (ICCV), 2017, pp. 2999–3007.
27. Ioffe, S.; Szegedy, C. Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. In Proceedings of the Proc. Int. Conf. Machine Learning (ICML), 2015, pp. 448–456.
28. Hendrycks, D.; Gimpel, K. Gaussian Error Linear Units (GELUs). *arXiv preprint arXiv:1606.08415* **2016**.
29. Srivastava, N.; Hinton, G.; Krizhevsky, A.; Sutskever, I.; Salakhutdinov, R. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine Learning Research* **2014**, *15*, 1929–1958.
30. He, K.; Zhang, X.; Ren, S.; Sun, J. Deep Residual Learning for Image Recognition. In Proceedings of the Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.
31. Wang, G.; Li, W.; Aertsen, M.; Deprest, J.; Ourselin, S.; Vercauteren, T. Aleatoric Uncertainty Estimation with Test-Time Augmentation for Medical Image Segmentation with Convolutional Neural Networks. *Neurocomputing* **2019**, *335*, 34–45.
32. Kingma, D.P.; Ba, J. Adam: A Method for Stochastic Optimization. In Proceedings of the Proc. Int. Conf. Learning Representations (ICLR), 2015.

33. Loshchilov, I.; Hutter, F. Decoupled Weight Decay Regularization. In Proceedings of the Proc. Int. Conf. Learning Representations (ICLR), 2019.
34. Smith, L.N.; Topin, N. Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates. In Proceedings of the Proc. SPIE 11006, Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications, 2019, p. 1100612. https://doi.org/10.1117/12.2520589.